

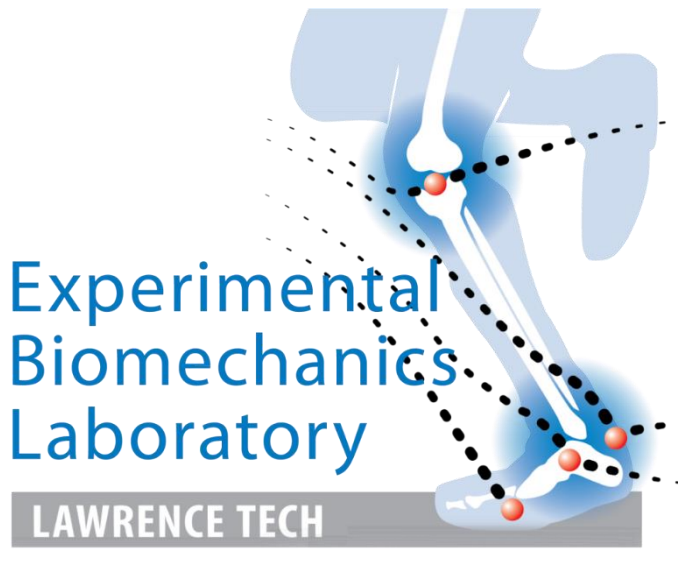
THE RISK OF RECURRENT DISC HERNIATION FOLLOWING DECOMPRESSION SURGERY WITH SURGICAL DETACHMENT OF THE POSTERIOR EXTENSOR MUSCLES

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INTRODUCTION

- The most common low-back surgery performed in the United States is lumbar discectomy, which often includes facetectomy and laminotomy [2,3].
- Recurrent disc herniations following surgical decompression has been reported in patients at rates between 5% and 21% and often require additional operations [4,5].
- The facet joints and lumbar posterior extensor muscles (PEMs) help maintain proper spinal biomechanics, and decrease the risk for recurrent herniation [6,7].
- Surgical detachment of the PEMs during decompression surgery may limit the spine's ability to fully engage the facet joints and result in an increased risk of recurrent herniation.
- OBJECTIVE: To evaluate the risk of herniation following bony decompression surgery with varying amounts of PEM detachment.**
- HYPOTHESIS:** Removal of the facet joint (i.e., facetectomy) or detachment of the PEMs would result in a disc loading environment consistent with an increased herniation risk.

METHODS

- Utilized a finite element (FE) model of a lumbar motion segment that has been demonstrated to accurately predict disc herniation risk.
- Boundary conditions were chosen to simulate forward bending for a 50th percentile male (Figure 1). The L5 inferior bony endplate was fixed rigidly in space.
- Surgically altered the FE model by either unilateral hemilaminotomy or unilateral facetectomy (Figure 2).
- PEM muscle detachment was simulated by reducing the muscle force.
- The following scenarios were simulated: unaltered; hemilaminotomy with no PEM detachment, partial unilateral PEM detachment, and complete unilateral PEM detachment; facetectomy with limited unilateral PEM detachment and complete unilateral PEM detachment.

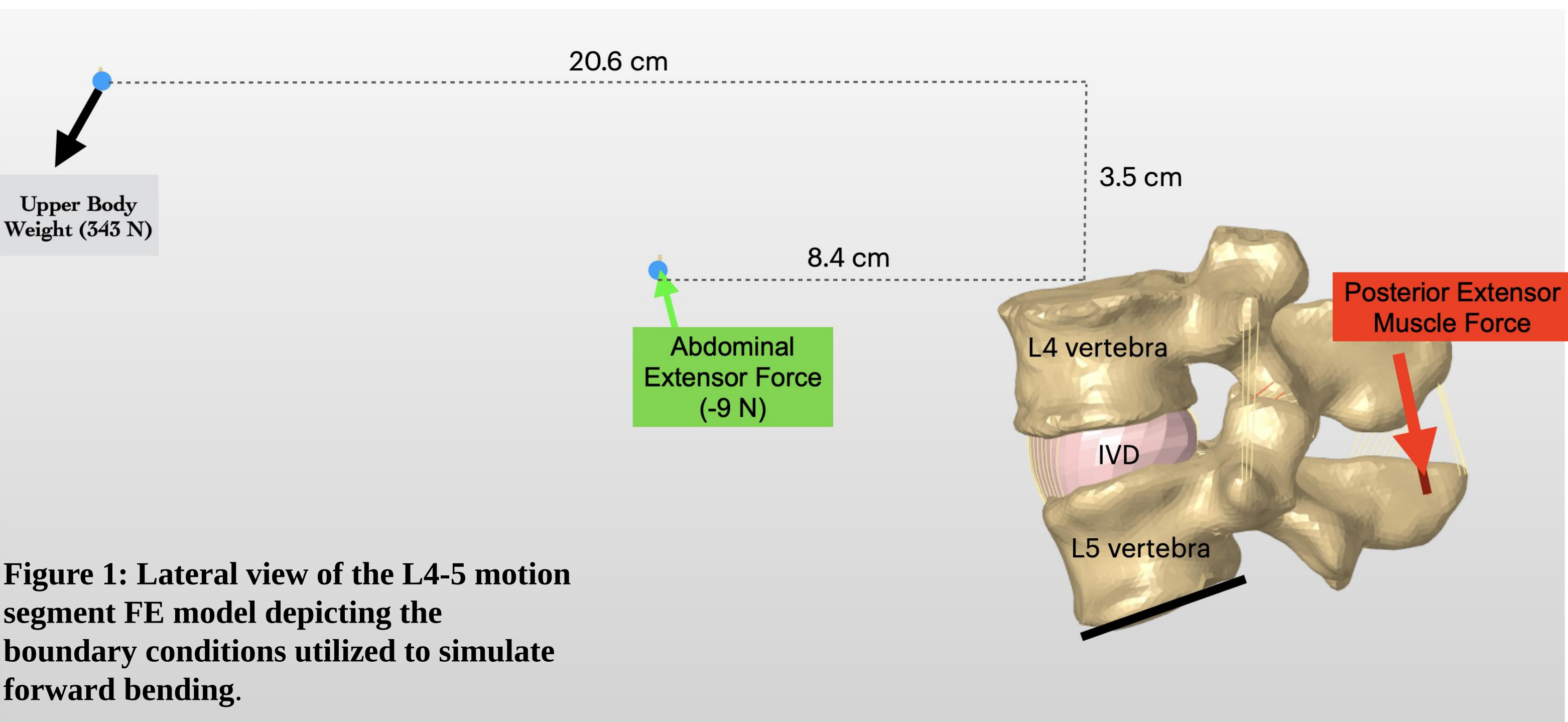


Figure 1: Lateral view of the L4-5 motion segment FE model depicting the boundary conditions utilized to simulate forward bending.

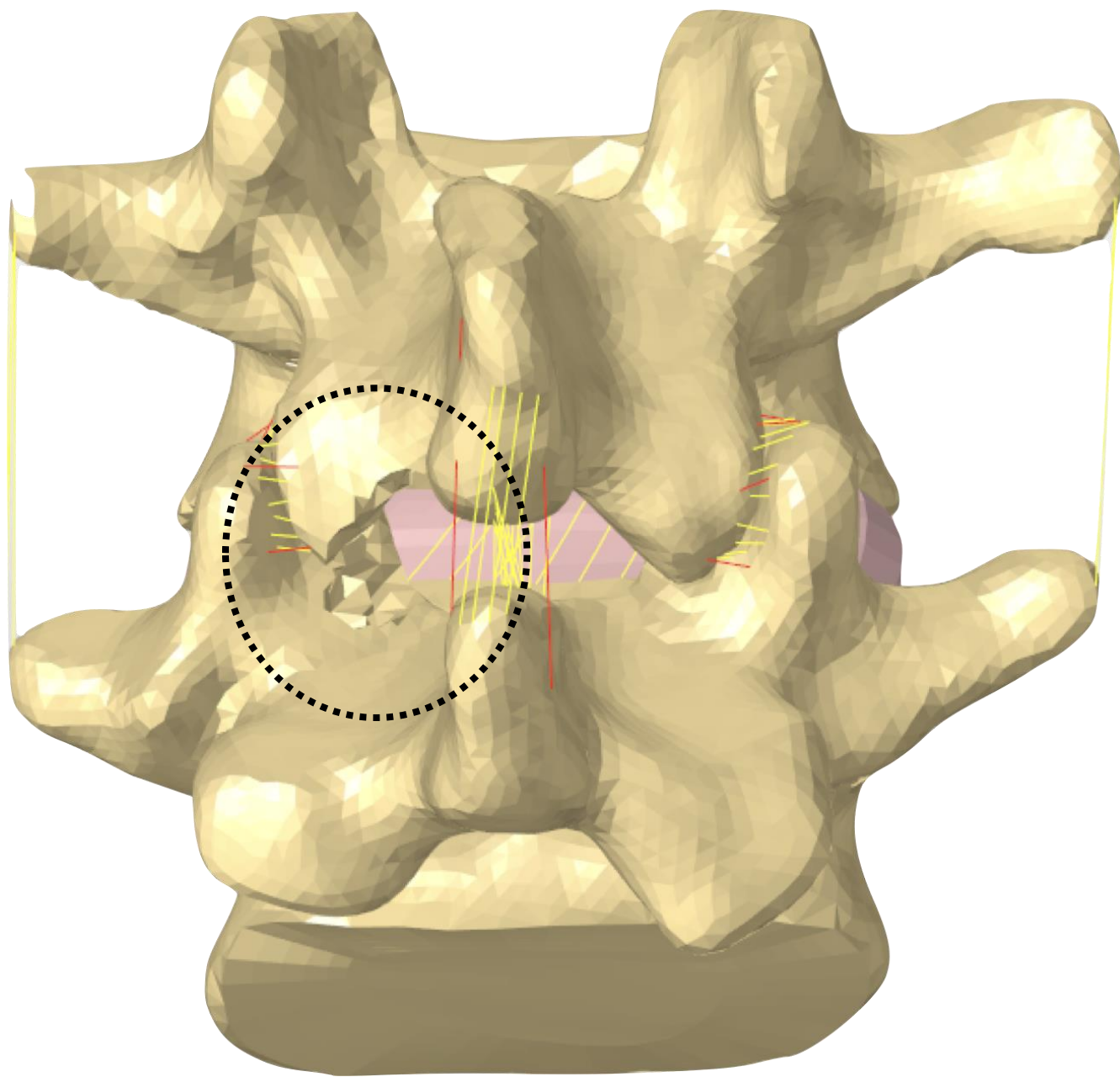
DISCUSSION

- Detachment of the PEM reduced facet joint contact force and increased disc stresses. However, removal of the facet joint did not increase disc stress. Rather, the greatest risk for recurrent herniation occurred because of PEM detachment and not bone removal.
- CONCLUSION: There was no significant change in the risk of recurrent herniation after hemilaminotomy or facetectomy when the muscles were left intact or when there was only limited muscle detachment.**
- Regardless of the surgical technique utilized to treat the disc herniation, clinical data and the results of the current study suggest that PEM damage leaves the disc most vulnerable to recurrent herniation.
- If removal of the laminae or facet joint is necessary, care should be taken to avoid detachment of the PEM from the vertebrae. However, if muscle detachment is unavoidable, post-surgical rehabilitation and care should be taken to minimize risk to the disc. Specifically, exercises that strengthen the remaining torso muscles and bending and lifting techniques that accentuate bending with a taught back should be emphasized. Additionally, establishing pre- and post-op methods for quantifying muscle strength may assist in determining an optimal recovery strategy.

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Left Hemilaminotomy



Right Hemilaminotomy

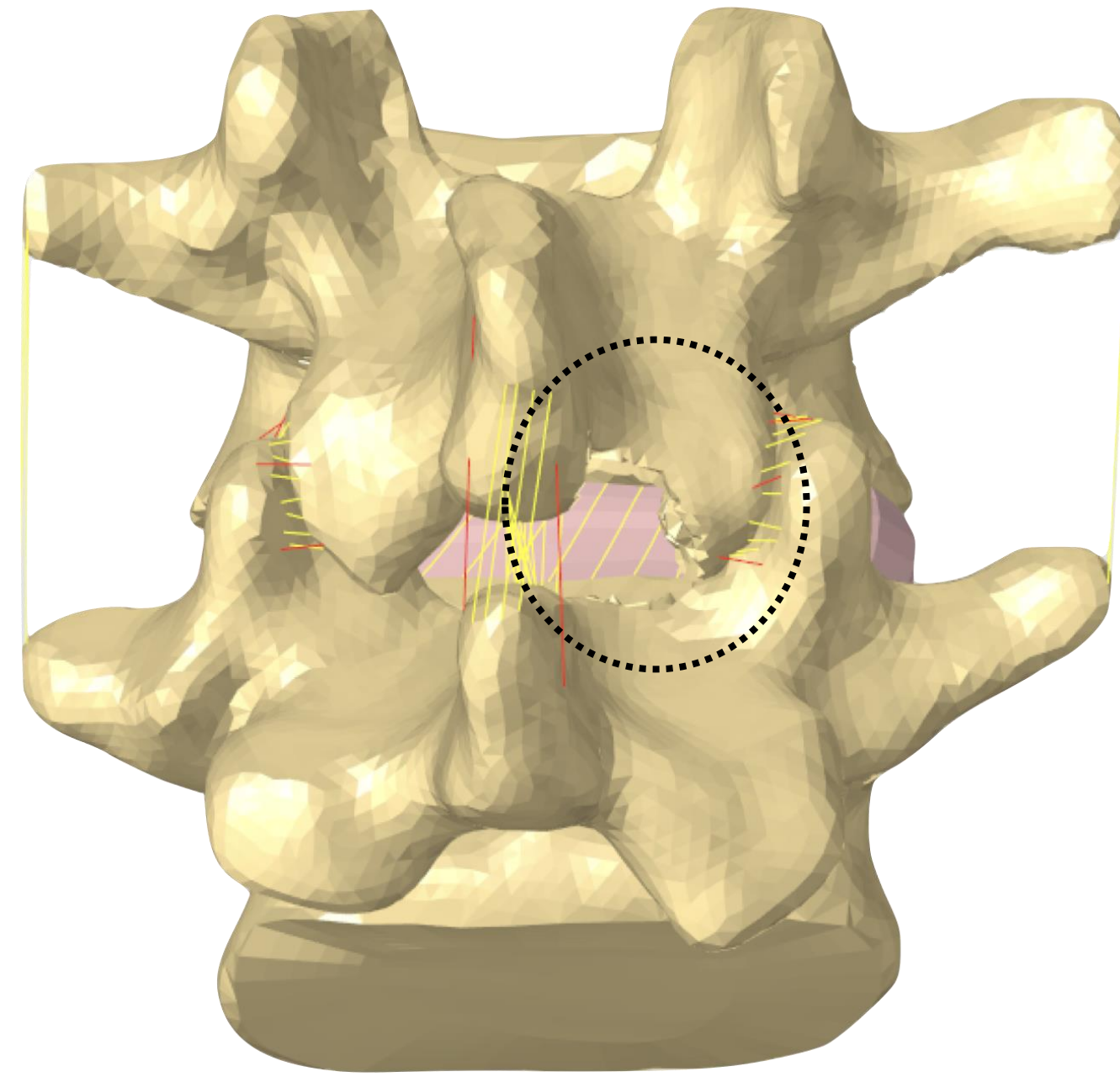
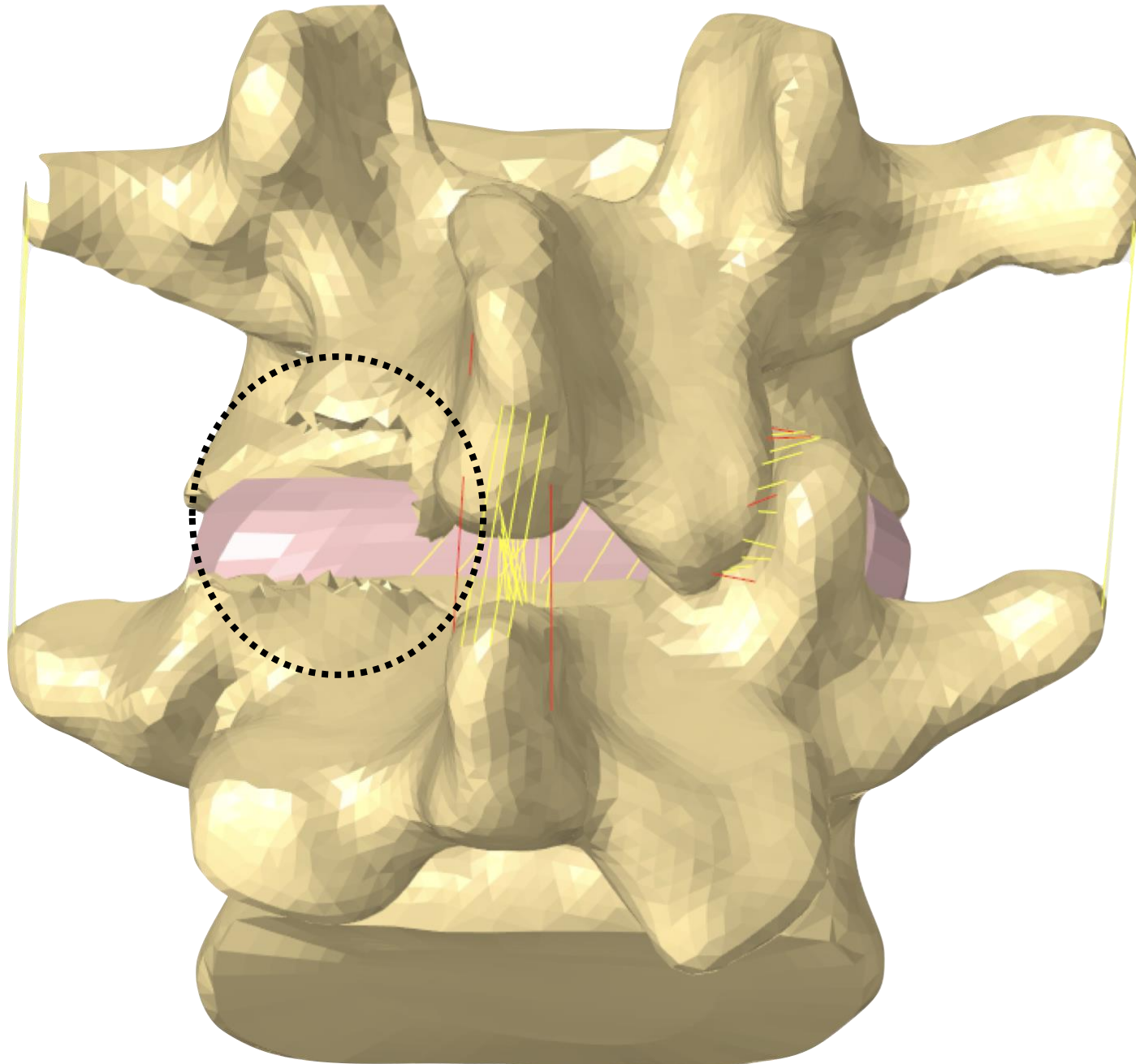
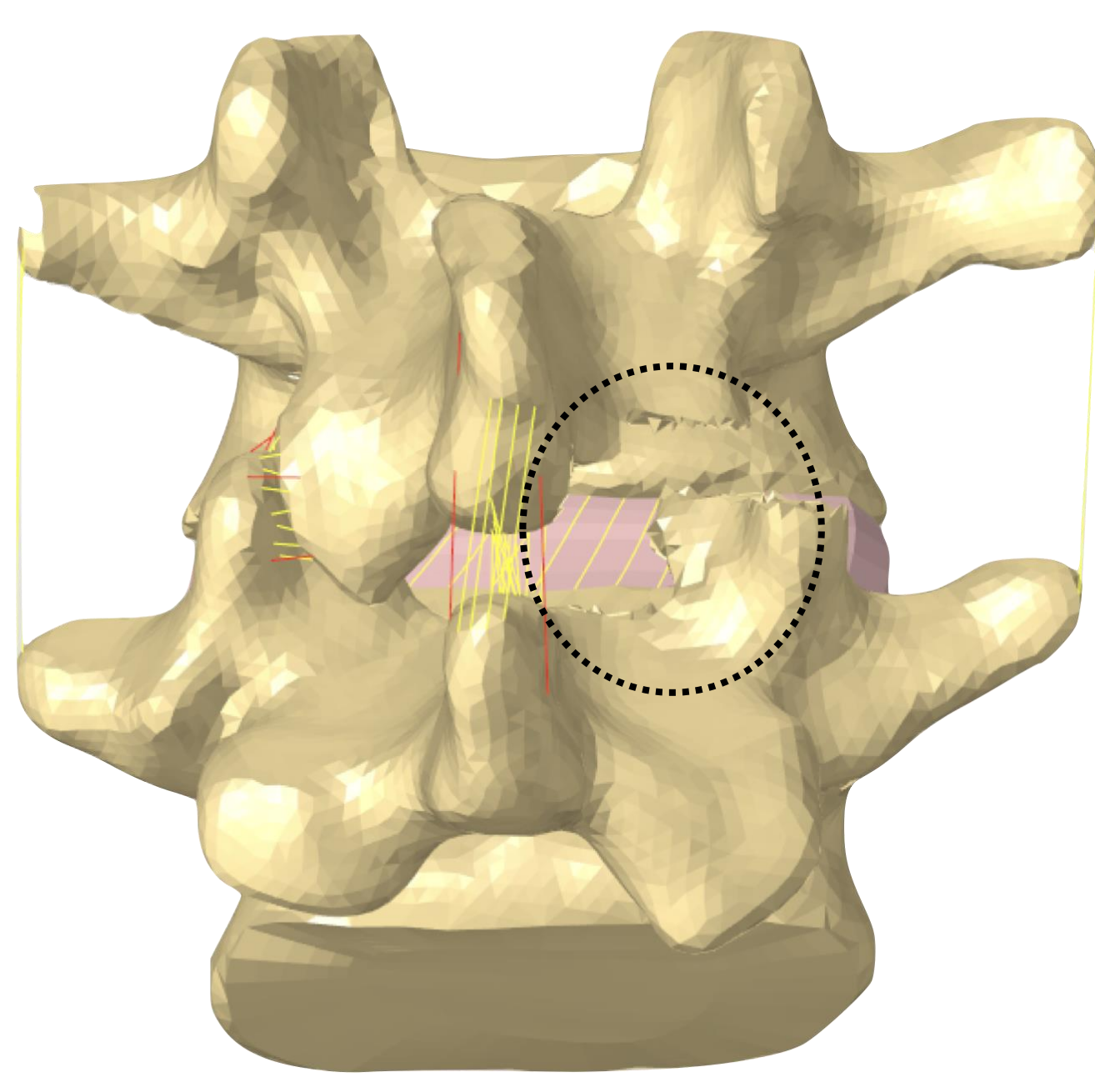


Figure 2: Posterior views of the surgically altered FE model. The locations of surgical alteration are depicted in the circled area.

Left Facetectomy



Right Facetectomy



RESULTS

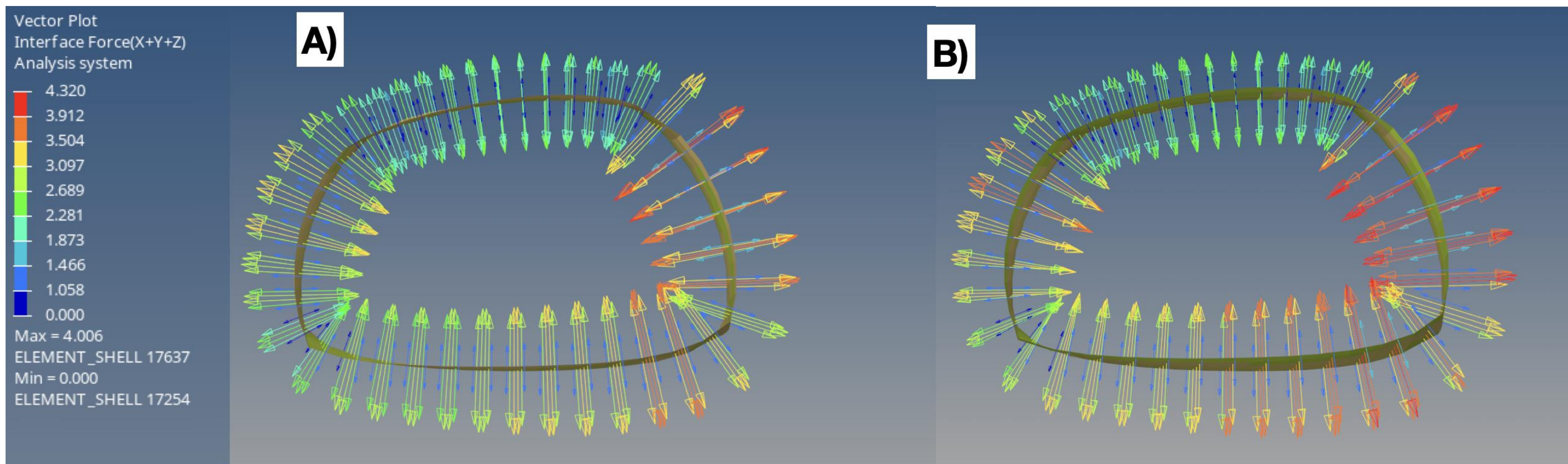


Figure 3: The interface contact force between the posterolateral circumference of the nucleus and inner annulus (i.e., NP extrusion force) ranged between 310 and 319 N for all scenarios with intact PEMs. The posterolateral NP extrusion force increased as the amount of PEM force was decreased. This image depicts the NP extrusion force of the A) unaltered model compared to B) complete PEM detachment after right hemilaminotomy.

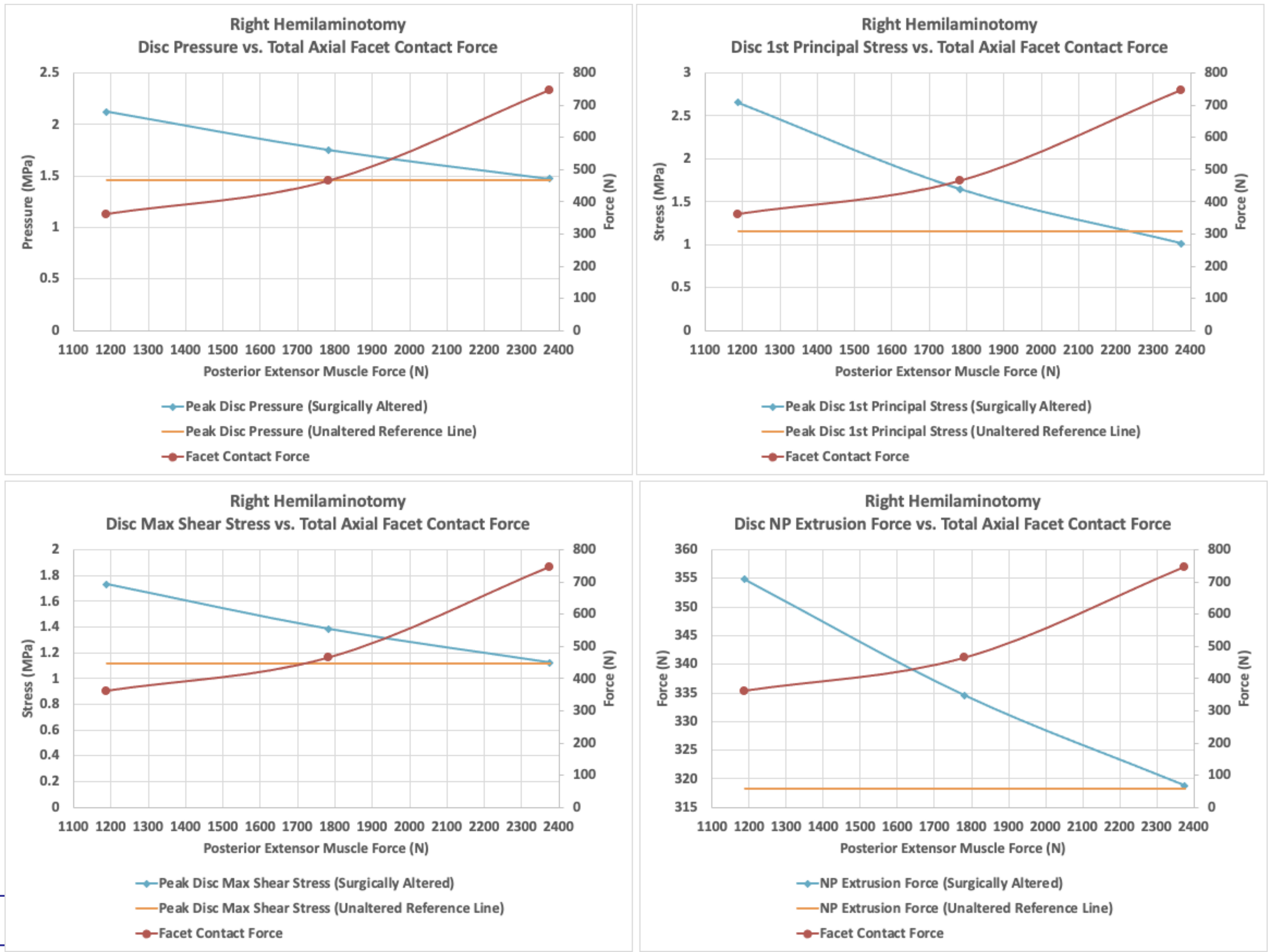


Figure 4: There was an inverse relationship between facet joint contact force and disc pressure/stresses as the PEM force decreased. Specifically, after right hemilaminotomy, the peak disc pressures increased from 1.47 MPa to 2.12 MPa, and the facet joint contact force decreased from 746 N to 361 N as the amount of PEM force decreased from 2,376 N to 1,188 N. When the PEM was left intact, there were insignificant differences between the unaltered model and the surgically altered model. Similar trends were seen for peak disc 1st principal stress, peak disc max shear stress, and posterolateral NP extrusion force. This image depicts line graphs showing the correlations between facet joint contact force, PEM force, and peak disc pressures, disc 1st principal stresses, disc max shear stresses, and NP extrusion forces following right hemilaminotomy.